

Deliberation and Learning: Evidence From Deliberative Polls*

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Abstract

We study factual learning among 8,800 participants in 592 discussion groups across 28 Deliberative Polls held between 1994 and 2021. The polls address elections, energy, health, crime, climate, and other public issues in Europe, Australia, China, and the United States. Averaging equally across polls, knowledge scores increased by 13.7 percentage points, and the average increase relative to each poll’s initial score was 37.7%. Adjustment for guessing yields estimated learning of 18.2 percentage points. In two polls with before-and-after control interviews, attendees’ gains exceeded controls’ gains by 13.1 and 9.3 percentage points; their average difference was 11.2 percentage points, or 14.0 percentage points after guessing adjustment. We assess uncertainty by resampling polls and their discussion groups.

1 Introduction

Among the great virtues commonly ascribed to deliberation is learning. The participants learn about the topics under discussion and, to a lesser degree, incidentally, about related ones. They learn not only relevant facts but the arguments for and against particular alternatives and what other people unlike themselves, their circumstances, and their ways of thinking are like. This is helpful, both at the individual level, for discerning what policies most closely accord with one’s values and interests and voting accordingly, and at the aggregate level, for forming “authentic” majorities—majorities that “know what they are doing” (Luskin 2003). How much citizens know shapes what they prefer and how they vote (Converse 1964; Bartels 1996; Althaus 1998; Delli Carpini and Keeter 1996), even on foreign policy (Aldrich et al. 1989; Foyle 2004), though information shortcuts can partly stand in for knowledge (Popkin 1991; Lupia 1994).

The expectation of learning follows directly from the nature of “deliberation,” conventionally construed as discussion, but not just any discussion (Fishkin and Luskin 2005). While definitions vary, deliberation is generally viewed as normatively elevated discussion—in which, among other things, the participants clarify and refine their preferences by open-mindedly weighing competing arguments. As part of that weighing, they absorb factual information and sift out misinformation.

Or so the theory runs. But how much do participants actually learn, is it the deliberation that teaches them, and who learns the most? We examine learning of relevant facts. This is not the only sort of learning involved, but it is the easiest to measure, and it is both important in itself and positively correlated with the rest.

We compare the knowledge participants bring to deliberation with what they know afterward, using one sample throughout the main analysis. Two of these polls also interviewed people who were not

*Data and replication materials: <https://github.com/soodoku/dp-learning>.

invited, allowing us to compare changes among attendees and controls. Earlier studies document learning in Deliberative Polls and other forums (Luskin et al. 2002; Barabas 2004; Sturgis et al. 2005; Esterling et al. 2011; Theuwis et al. 2025). We report gains in both absolute and relative terms, adjust for guessing, and examine whether participants learn more alongside more knowledgeable groupmates.

2 Deliberative Polling and Learning

Deliberative Polling (Fishkin 1991, 1995; Luskin et al. 2002) aims to capture fuller-information, more thoughtful opinions than those expressed in ordinary polls. In very brief, a Deliberative Poll interviews and assembles, as many as are willing of, a random sample to deliberate some policy or electoral choice, in alternating small group discussions and plenary question-and-answer sessions with panels of competing experts, and then re-measures the sample’s opinions at the end. These latter are frequently different, both at the individual level and in the aggregate, from what they were before deliberating. The absolute net change is statistically significant nearly three-quarters of the time and averages nearly .1 on a 0–1 scale (Luskin et al. 2007).

This design, admittedly a particular embodiment of deliberation, affords additional opportunities and incentives to learn. The participants are provided with carefully balanced briefing documents, rehearsing the arguments each side would make. There is usually a section of basic background information, and the arguments embed each side’s pet facts—those most congenial to its arguments. In addition, the participants know from the moment they are first recruited that they will be attending the deliberation and that the deliberation (typically) will be an “Event,” capital “e”—away from home, usually televised or taped for later broadcast, often attended by high-profile policy experts or policy-makers in the role of observers or panelists. Thus, motivated by a desire to “do well,” many participants use the weeks between their recruitment and the event itself to prepare—not only by reading the briefing materials but by paying heightened attention to stories about the topic in the media, by talking more with friends, family, and coworkers about it, and even, in some cases, by researching it in the library or online.

So if knowledge, like its close relative “sophistication,” depends on the confluence of cognitive ability, motivation, and exposure to relevant information (Luskin 1990, 2002), the deliberation in Deliberative Polling stands to increase two of those three. There is not much to be done, in adulthood, about ability—and in fact we should want the participants to be representative in this, as in all other respects. But the treatment can and does affect motivation and exposure. Judging from past polls, that is enough to produce considerable, sometimes dramatic learning (Luskin et al. 2002; List et al. 2013).

The online design is intended to be as similar as possible to the face-to-face one, given the difference in mode. The deliberation is by voice rather than text, so as to capture as much as possible of deliberation’s social elements. The participants still “meet” in small groups, still led by trained moderators, and still question panels of policy experts and policy makers. They still receive briefing materials. They are still a representative sample—sometimes randomly sampled offline, with “nonusers” receiving free access, including free equipment, to get them online, sometimes drawn from online panels and matched to approximate a random sample. On the other hand, deliberating by voice, even by video, online can never capture the full richness and complexity of in-person interactions; the discussions are interrupted, spread over a period of weeks, with much greater possibility of real-world intrusion, and the assignment to small groups is perforce constrained by possible meeting times. All these differences may act to weaken the online treatment’s effects

(Luskin et al. 2006; Iyengar et al. 2005).

3 Data

The main sample comprises 28 polls and 8,800 attendees with knowledge answers before and after deliberation and recorded discussion-group membership. It supports both the learning estimates and the analysis of discussion groups. We use the versioned Deliberative Poll dataset (Sood 2026), selecting one respondent record per poll and averaging item correctness into knowledge scores. Appendix C gives the questions and scoring rules.

America in One Room and the climate poll interviewed separate samples who were never invited to deliberate (Fishkin et al. 2021). We compare changes in their scores with changes among attendees in the main sample, using the same questions in both arms. These comparisons may reflect selection into attendance as well as deliberation. Additional control comparisons and follow-ups appear in Appendix B.4.

4 Measuring Learning

We measure knowledge by closed-ended knowledge questions, mostly multiple choice, in some cases extracted from party or candidate placements, and in a few cases true-false (Luskin 1987; Luskin and Bullock 2005; Price 1999; Mondak 1999, 2001). Appendix C lists the items. Correct responses are scored as 1, incorrect responses and “don’t knows” as 0 (Luskin and Bullock 2011), and these scores are then averaged across items to give the proportion answered correctly.

Table 1 models the change in proportion correct, $K_2 - K_1$, conditional on initial knowledge K_1 . It compares observed gains among participants with the same measured starting score. Including K_1 makes this algebraically equivalent to modeling K_2 conditional on K_1 , as Section 6.2 shows. An unadjusted change-score model instead constrains the baseline coefficient to zero in the gain equation. These specifications can disagree when groups differ at baseline (Allison 1990; van Breukelen 2006). Score ceilings, guessing, and measurement error complicate the relationship between observed gains and learning of the underlying facts (Luskin et al. 2016). We therefore treat the T1 coefficient as an association, not an estimate of how prior knowledge causes learning.

A correct answer can reflect knowledge or guessing. We apply the item model of Cor and Sood (2016) separately to each poll’s paired answers. Under the assumption that knowledge is not lost between the two interviews, it estimates the share who learned each fact. We average across questions to obtain learning on a 0–1 scale. In each control study, we fit this model separately to attendees and controls and subtract their estimated learning. Both arms use the same items; their guessing probabilities are estimated separately. Table 1 uses observed individual gains, since the item model estimates learning rates rather than an individual latent outcome.

We estimate uncertainty using a hierarchical bootstrap. For estimates across polls, we sample polls with replacement separately within online and in-person modes, retaining their observed counts, and then sample whole discussion groups within each selected poll. For an individual poll, we hold the poll fixed and sample its groups. Independently sampled controls are single-person units, sampled separately from attendees. Each sampled participant retains all items and interviews. We refit the models and recompute denominators in each of 999 draws, using the 2.5th and 97.5th percentiles for 95% confidence intervals. Thus the same resampling scheme propagates sampling

and guessing-model estimation uncertainty. Stratifying by mode preserves support for the online coefficient in every draw. Inference treats each poll’s question battery as fixed.

5 How Much Do Participants Learn?

An absolute gain is the change in the percentage of items answered correctly. A relative gain divides that change by the poll’s initial mean score: an increase from 40% to 50% correct is 10 percentage points, or 25% of the initial score. We calculate relative gains from poll means rather than averaging individual ratios, which are undefined for people with zero initial scores. The main pooled estimates give each poll equal weight; the relative result is the average of the poll-specific ratios, not the ratio of two pooled means.

Across the main sample, the average observed gain is 13.7 percentage points (95% interval 11.4 to 16.2). The average relative increase is 37.7% (95% interval 28.2 to 49.7%). Estimated learning after adjustment for guessing averages 18.2 percentage points (15.7 to 21.4). Appendix Table A3 also reports precision-weighted averages and separate estimates by mode.

A before-and-after gain can include learning from repeated questions or events outside deliberation. Figure 1 therefore also compares attendees’ gains with controls’ gains. The observed differences are 13.1 percentage points in America in One Room and 9.3 percentage points in the climate poll. Their equally weighted average is 11.2 percentage points (95% interval 9.7 to 12.7), rising to 14.0 percentage points after guessing adjustment (12.2 to 15.7). Both estimates use the same attendees, controls, and questions. These comparisons do not eliminate selection into attendance.

6 Who Learns?

6.1 Regressors and hypotheses

Table 1 begins with initial knowledge, groupmates’ initial knowledge, group size, mode, and the poll’s mean initial score. Its second column adds education, age, gender, and the gender composition of the group. These variables are available in all 28 polls. Education may improve the acquisition of new information (Luskin 1990); we allow its association with gain to vary with initial knowledge. The interaction between gender and the proportion of female groupmates tests whether that association varies with group composition. Appendix B develops the additional hypotheses about attitudes and briefing reading.

Initial knowledge (K_1). The information-rich tend to get information-richer. The more one knows going in, the more one tends to learn and thus know going out. Absent ceiling effects, this predicts a positive association with gain; bounded scores can instead produce a negative association. Prior knowledge on a particular topic improves people’s ability to comprehend new information, enabling them to extract the central elements of a message and draw appropriate inferences efficiently (Eckhardt et al. 1991; Recht and Leslie 1988). Prior knowledge also enhances people’s ability to store new information on that topic and retrieve it later (Cooke et al. 1993; Fiske et al. 1990; Hambrick 2003; McGraw and Pinney 1990; Recht and Leslie 1988; Schneider et al. 1993). So the less political information individuals have stored in memory, the more difficult it is for them to acquire new information.

Mode. Online deliberation, with its thinner exposure to other participants, may be, either intrinsically or at least in its current instantiations, a somewhat etiolated version of the face-to-face

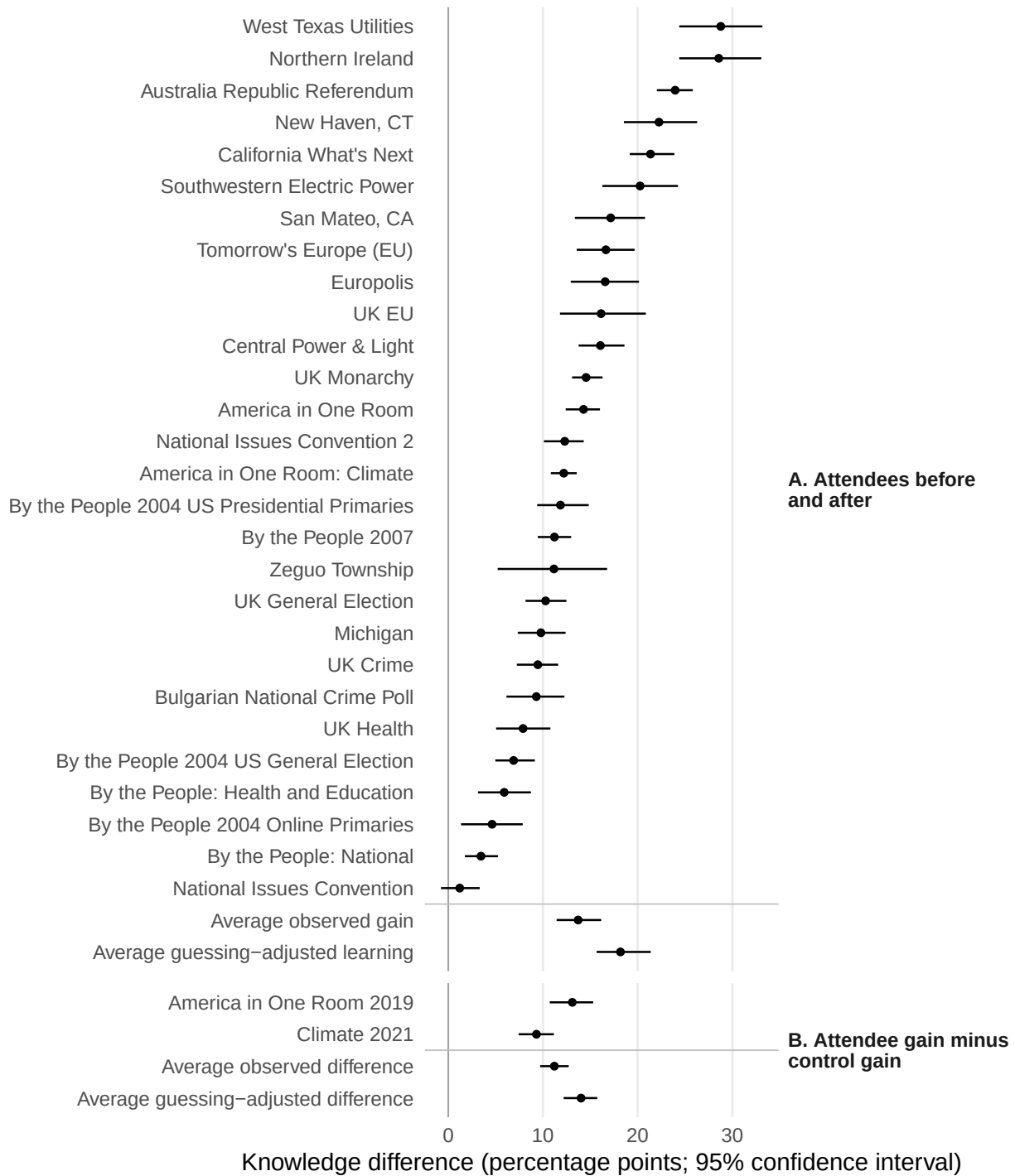


Figure 1: Learning in percentage points, with 95% hierarchical-bootstrap confidence intervals. A: observed attendee gains in the main-sample polls. B: observed attendee gain minus control gain in the two polls with common baseline and follow-up interviews. Each panel ends with equally weighted observed and guessing-adjusted averages. Whole discussion groups are resampled within polls; independent controls are resampled individually. Pooled intervals resample polls within mode. With one control study per mode, the pooled control interval is conditional on these two studies. Each draw refits the guessing model, treating question batteries as fixed.

treatment (Luskin et al. 2006). It is therefore plausible that people learn less from online deliberations.

Group size. We count group members with both baseline and follow-up answers. This is a lower bound on actual group size when a member lacks one of the interviews. Larger groups may offer more perspectives but give each participant less time to speak (Latané et al. 1979).

Group knowledge. Participants may learn more from the other members of their small group when the latter know more. The other members’ mean T1 score measures their overall knowledge. Their mean correctness on the particular items a participant missed at T1 measures knowledge that could fill that participant’s gaps. We add this targeted measure alongside the overall group mean in a supplementary model. Participants who missed no T1 items receive zero on the targeted measure and a separate indicator for having no missed items.

Initial knowledge in the poll. We include the poll’s mean T1 score to account for differences in baseline knowledge across events. It reflects both attendees’ knowledge and the difficulty of the question battery. It is a contextual control; we do not interpret it as a direct measure of issue salience.

6.2 Model

Let $G_{igd} = K_{igd2} - K_{igd1}$ denote the change in proportion correct for participant i in discussion group g and poll d . We fit a hierarchical linear model (Raudenbush and Bryk 2002):

$$G_{igd} = \beta_0 + \beta_1 K_{igd1} + \mathbf{z}'_{igd} \beta_z + \alpha_{gd} + \gamma_d + \varepsilon_{igd},$$

where $\mathbf{z}_{igd}$ contains groupmates’ mean initial knowledge, group size, mode, and the poll’s mean initial score. The second specification adds education, age, gender, the proportion of female groupmates, and the education-by-T1 and gender-by-group-composition interactions. The β coefficients are fixed regression parameters. For fitting, we assume

$$\alpha_{gd} \sim N(0, \sigma_g^2), \quad \gamma_d \sim N(0, \sigma_p^2), \quad \varepsilon_{igd} \sim N(0, \sigma_e^2),$$

with these components mutually independent conditional on the regressors and independent across distinct groups, polls, and individuals, respectively. The group and poll random intercepts allow outcomes to covary within discussion groups and across groups in the same poll. Their variances, σ_g^2 and σ_p^2 , measure residual variation at those levels; σ_e^2 measures individual residual variation. We estimate the coefficients and variance components by maximum likelihood (Bates et al. 2015) and obtain confidence intervals from the hierarchical bootstrap.

Adding K_{igd1} to both sides gives the equivalent post-deliberation equation:

$$K_{igd2} = \beta_0 + (1 + \beta_1) K_{igd1} + \mathbf{z}'_{igd} \beta_z + \alpha_{gd} + \gamma_d + \varepsilon_{igd}.$$

Thus using gain instead of T2 shifts the coefficient on T1 down by one and leaves the other coefficients and fitted residuals unchanged. We report gain because it directly expresses the change the paper seeks to explain. Conditioning on T1 compares that change among participants with the same measured starting knowledge.

The analysis includes 8,800 participants in 592 groups and 28 polls. The demographic model retains 8,671 participants and all 28 polls; 129 participants have missing covariates. Appendix Table A6 distinguishes changes due to sample restrictions from changes due to added controls.

Table 1: Change in proportion correct (T2 minus T1), conditional on T1, in 28 polls. Coefficients with 95 percent hierarchical-bootstrap confidence intervals.

	Baseline	Demographic controls
Intercept	.194 [.128, .266]	.070 [-.026, .164]
T1 knowledge	-.490 [-.553, -.436]	-.518 [-.620, -.402]
Secondary/some college		.093 [.045, .151]
BA or more		.192 [.141, .254]
T1 x secondary/some college		.010 [-.103, .116]
T1 x BA or more		-.066 [-.183, .039]
Age (decades)		.002 [-.004, .008]
Group size	-.001 [-.003, .001]	-.001 [-.002, .001]
Groupmates' mean T1	.048 [-.086, .198]	.030 [-.095, .156]
Female		-.021 [-.071, .030]
Group share women		-.014 [-.078, .044]
Female x group share women		.017 [-.079, .113]
Online	-.067 [-.109, -.017]	-.089 [-.130, -.041]
Poll's mean initial score	.386 [.183, .584]	.492 [.279, .695]
Participants	8,800	8,671
Small groups	592	592
Polls	28	28
Marginal R-squared	0.234	0.272
Conditional R-squared	0.298	0.335

Notes. Groupmates' mean T1 excludes the respondent herself; both that mean and group size use members with baseline and follow-up answers. Both models have poll and small-group random intercepts. Education is coded as below secondary completion, secondary/some college, or degree or higher; the first category is the reference. Marginal R^2 covers fixed effects; conditional R^2 includes random intercepts. The models with additional attitude measures and reading reports appear in Appendix Table A5.

6.3 Who gains more?

Table 1 compares gains conditional on measured initial knowledge. The T1 coefficient in the baseline model is -.490 [-.553, -.436]. Because scores have a ceiling and contain measurement error, this should not be read as a causal effect of prior knowledge. The second column brings the education, age, and gender hypotheses into the same set of polls. Appendix Table A5 adds attitude extremity and opinion heterogeneity in 21 polls and examines briefing-material reading in 12 polls. Appendix Table A6 fits the baseline specification again on each restricted sample before adding covariates, making the source of any coefficient change explicit. Appendix Table A4 examines whether attendee-control differences vary with initial knowledge or education.

At the reference initial score of zero, both education categories have larger estimated gains than the below-secondary group; the education-by-T1 interactions allow these gaps to vary with starting knowledge. The age estimate is .002 [-.004, .008] per decade. The intervals for age, gender, and the gender-by-group-composition interaction include zero. The gender coefficient compares women with men when the share of female groupmates is zero; the interaction shows how that difference changes as the share increases.

Reported reading of the briefing materials is also associated with larger gains. Across 12 polls, the

difference between the lowest and highest reading categories is 6.3 percentage points (95% interval 3.2 to 9.5), conditional on initial knowledge, demographics, and poll fixed effects. These reports were collected after deliberation, so the association need not reflect the effect of reading alone.

6.4 Group composition and learning

Table 1 addresses the peer-knowledge hypothesis directly. A ten-percentage-point difference in groupmates' mean initial knowledge is associated with 0.48 percentage points of additional gain in the baseline model and 0.30 after demographic adjustment. The intervals in the table quantify uncertainty about these pooled associations.

Appendix Figure A1 estimates the association separately within each poll. It shows whether the pooled result conceals different patterns across settings. Its average weights those separate slopes equally; Table 1 instead fits a common slope in a hierarchical model. Its contribution is to display variation across polls.

Appendix Table A7 examines whether participants with similar baseline characteristics were grouped together. For initial knowledge, the association with groupmates' initial knowledge is .071 [-.026, .166]. Its interval includes zero. This diagnostic concerns characteristics measured before discussion; T2 and gain can themselves respond to group composition. A causal interpretation of peer coefficients also requires evidence about the assignment process. Group knowledge after deliberation is difficult to interpret because groupmates share the same experience (Manski 1993).

7 Conclusion

Participants' factual knowledge increases across the polls we study. Reporting absolute gains in percentage points and increases relative to initial scores makes the magnitude interpretable, while adjustment for guessing addresses one source of measurement error. In the two polls with common before-and-after control interviews, attendees show larger gains than controls. The comparisons remain subject to selection into attendance, and the peer regressions are descriptive. The knowledge questions concern facts selected by organizers, so these estimates describe learning about the topics the events emphasized.

References

- Aldrich, John H., John L. Sullivan, and Eugene Borgida. 1989. "Foreign Affairs and Issue Voting: Do Presidential Candidates 'Waltz Before a Blind Audience?'" *American Political Science Review* 83 (1): 123–41. <https://doi.org/10.2307/1956437>.
- Allison, Paul D. 1990. "Change Scores as Dependent Variables in Regression Analysis." *Sociological Methodology* 20: 93–114. <https://doi.org/10.2307/271083>.
- Althaus, Scott L. 1998. "Information Effects in Collective Preferences." *American Political Science Review* 92 (3): 545–58. <https://doi.org/10.2307/2585480>.
- Anderson, T. W. 2003. *An Introduction to Multivariate Statistical Analysis*. 3rd ed. Wiley-Interscience.

- Barabas, Jason. 2004. "How Deliberation Affects Policy Opinions." *American Political Science Review* 98 (4): 687–701. <https://doi.org/10.1017/s0003055404041425>.
- Bartels, Larry M. 1996. "Uninformed Votes: Information Effects in Presidential Elections." *American Journal of Political Science* 40 (1): 194–230. <https://doi.org/10.2307/2111700>.
- Bates, Douglas, Martin Mächler, Ben Bolker, and Steve Walker. 2015. "Fitting Linear Mixed-Effects Models Using lme4." *Journal of Statistical Software* 67 (1). <https://doi.org/10.18637/jss.v067.i01>.
- Converse, Philip E. 1964. "The Nature of Belief Systems in Mass Publics." In *Ideology and Discontent*, edited by David E. Apter. Free Press.
- Cooke, Nancy J., Robert S. Atlas, David M. Lane, and Robert C. Berger. 1993. "Role of High-Level Knowledge in Memory for Chess Positions." *The American Journal of Psychology* 106 (3): 321–51. <https://doi.org/10.2307/1423181>.
- Cor, M. Ken, and Gaurav Sood. 2016. "Guessing and Forgetting: A Latent Class Model for Measuring Learning." *Political Analysis* 24 (2): 226–42. <https://doi.org/10.1093/pan/mpw010>.
- Delli Carpini, Michael X., and Scott Keeter. 1996. *What Americans Know about Politics and Why It Matters*. Yale University Press.
- Eckhardt, Beverly B., Mary R. Wood, and Robin Smith Jacobvitz. 1991. "Verbal Ability and Prior Knowledge: Contributions to Adults' Comprehension of Television." *Communication Research* 18 (5): 636–49. <https://doi.org/10.1177/009365091018005004>.
- Esterling, K. M., M. A. Neblo, and D. M. J. Lazer. 2011. "Means, Motive, and Opportunity in Becoming Informed about Politics: A Deliberative Field Experiment with Members of Congress and Their Constituents." *Public Opinion Quarterly* 75 (3): 483–503. <https://doi.org/10.1093/poq/nfr001>.
- Fishkin, James S. 1991. *Democracy and Deliberation: New Directions for Democratic Reform*. Yale University Press.
- Fishkin, James S. 1995. *The Voice of the People: Public Opinion and Democracy*. Yale University Press.
- Fishkin, James S., and Robert C. Luskin. 2005. "Experimenting with a Democratic Ideal: Deliberative Polling and Public Opinion." *Acta Politica* 40 (3): 284–98. <https://doi.org/10.1057/palgrave.ap.5500121>.
- Fishkin, James, Alice Siu, Larry Diamond, and Norman Bradburn. 2021. "Is Deliberation an Antidote to Extreme Partisan Polarization? Reflections on 'America in One Room'." *American Political Science Review* 115 (4): 1464–81. <https://doi.org/10.1017/s0003055421000642>.
- Fiske, Susan T., Richard R. Lau, and Richard A. Smith. 1990. "On the Varieties and Utilities of Political Expertise." *Social Cognition* 8 (1): 31–48. <https://doi.org/10.1521/soco.1990.8.1.31>.

- Foyle, Douglas C. 2004. "Leading the Public to War? The Influence of American Public Opinion on the Bush Administration's Decision to Go to War in Iraq." *International Journal of Public Opinion Research* 16 (3): 269–94. <https://doi.org/10.1093/ijpor/edh025>.
- Guryan, Jonathan, Kory Kroft, and Matthew J Notowidigdo. 2009. "Peer Effects in the Workplace: Evidence from Random Groupings in Professional Golf Tournaments." *American Economic Journal: Applied Economics* 1 (4): 34–68. <https://doi.org/10.1257/app.1.4.34>.
- Hambrick, David Z. 2003. "Why Are Some People More Knowledgeable Than Others? A Longitudinal Study of Knowledge Acquisition." *Memory & Cognition* 31 (6): 902–17. <https://doi.org/10.3758/bf03196444>.
- Hansen, Kasper M. 2004. *Deliberative Democracy and Opinion Formation*. University Press of Southern Denmark. <http://www.kaspermhansen.eu/Work/Hansen2004.pdf>.
- Iyengar, Shanto, Robert C. Luskin, and James S. Fishkin. 2005. *Deliberative Preferences in the Presidential Nomination Campaign: Evidence from an Online Deliberative Poll*. Working paper, Center for Deliberative Democracy, Stanford University. <http://cdd.stanford.edu/research/papers/2005/presidential-nomination.pdf>.
- Latané, Bibb, Kipling Williams, and Stephen Harkins. 1979. "Many Hands Make Light the Work: The Causes and Consequences of Social Loafing." *Journal of Personality and Social Psychology* 37 (6): 822–32. <https://doi.org/10.1037/0022-3514.37.6.822>.
- List, Christian, Robert C. Luskin, James S. Fishkin, and Iain McLean. 2013. "Deliberation, Single-Peakedness, and the Possibility of Meaningful Democracy: Evidence from Deliberative Polls." *The Journal of Politics* 75 (1): 80–95. <https://doi.org/10.1017/s0022381612000886>.
- Lupia, Arthur. 1994. "Shortcuts Versus Encyclopedias: Information and Voting Behavior in California Insurance Reform Elections." *American Political Science Review* 88 (1): 63–76. <https://doi.org/10.2307/2944882>.
- Luskin, Robert C. 1987. "Measuring Political Sophistication." *American Journal of Political Science* 31 (4): 856–99. <https://doi.org/10.2307/2111227>.
- Luskin, Robert C. 1990. "Explaining Political Sophistication." *Political Behavior* 12 (4): 331–61. <https://doi.org/10.1007/bf00992793>.
- Luskin, Robert C. 2002. "From Denial to Extenuation (and Finally Beyond): Political Sophistication and Citizen Performance." In *Thinking about Political Psychology*, edited by James H. Kuklinski. Cambridge University Press. <https://doi.org/10.1017/cbo9780511510632.010>.
- Luskin, Robert C. 2003. "The Heavenly Public: What Would a Fully Informed Citizenry Be Like?" In *Electoral Democracy*, edited by Michael B. MacKuen and George Rabinowitz. University of Michigan Press.
- Luskin, Robert C., and John G. Bullock. 2005. *Re(:)measuring Political Sophistication*. Paper presented at the annual meeting of the Midwest Political Science Association, Chicago. <https://doi.org/10.1017/cbo9780511510632.010>.

//web.stanford.edu/class/polisci353/2004fall/reading/BullockPiece.doc.

- Luskin, Robert C., and John G. Bullock. 2011. “‘Don’t Know’ Means ‘Don’t Know’: DK Responses and the Public’s Level of Political Knowledge.” *The Journal of Politics* 73 (2): 547–57. <https://doi.org/10.1017/s0022381611000132>.
- Luskin, Robert C., James S. Fishkin, and Kyu S. Hahn. 2007. *Deliberation and Net Attitude Change*. Paper presented at the ECPR General Conference, Pisa, Italy, September 6–8. <https://cdd.stanford.edu/mm/2007/deliberation-net-change.pdf>.
- Luskin, Robert C., James S. Fishkin, and Shanto Iyengar. 2006. *Considered Opinions on U.S. Foreign Policy: Evidence from Online and Face-to-Face Deliberative Polling*. Working paper, Center for Deliberative Democracy, Stanford University. <http://cdd.stanford.edu/research/papers/2006/foreign-policy.pdf>.
- Luskin, Robert C., James S. Fishkin, and Roger Jowell. 2002. “Considered Opinions: Deliberative Polling in Britain.” *British Journal of Political Science* 32 (3): 455–87. <https://doi.org/10.1017/s0007123402000194>.
- Luskin, Robert C., Ariel Helfer, and Gaurav Sood. 2016. “Measuring Learning in Informative Processes.” https://www.gsood.com/research/papers/k_gains.pdf.
- Manski, Charles F. 1993. “Identification of Endogenous Social Effects: The Reflection Problem.” *The Review of Economic Studies* 60 (3): 531. <https://doi.org/10.2307/2298123>.
- McGraw, Kathleen M., and Neil Pinney. 1990. “The Effects of General and Domain-Specific Expertise on Political Memory and Judgment.” *Social Cognition* 8 (1): 9–30. <https://doi.org/10.1521/soco.1990.8.1.9>.
- Mondak, Jeffery J. 1999. “Reconsidering the Measurement of Political Knowledge.” *Political Analysis* 8 (1): 57–82. <https://doi.org/10.1093/oxfordjournals.pan.a029805>.
- Mondak, Jeffery J. 2001. “Developing Valid Knowledge Scales.” *American Journal of Political Science* 45 (1): 224–38. <https://doi.org/10.2307/2669369>.
- Popkin, Samuel L. 1991. *The Reasoning Voter: Communication and Persuasion in Presidential Campaigns*. University of Chicago Press. <https://doi.org/10.7208/chicago/9780226772875.001.0001>.
- Price, Vincent. 1999. “Political Information.” In *Measures of Political Attitudes*, edited by John P. Robinson, Phillip R. Shaver, and Lawrence S. Wrightsman. Academic Press.
- Raudenbush, Stephen W., and Anthony S. Bryk. 2002. *Hierarchical Linear Models: Applications and Data Analysis Methods*. 2nd ed. Sage.
- Recht, Donna R., and Lauren Leslie. 1988. “Effect of Prior Knowledge on Good and Poor Readers’ Memory of Text.” *Journal of Educational Psychology* 80 (1): 16–20. <https://doi.org/10.1037/0022-0663.80.1.16>.

- Schneider, Wolfgang, Hans Gruber, Andreas Gold, and Klaus Opwis. 1993. “Chess Expertise and Memory for Chess Positions in Children and Adults.” *Journal of Experimental Child Psychology* 56 (3): 328–49. <https://doi.org/10.1006/jecp.1993.1038>.
- Sood, Gaurav. 2026. *Deliberative Poll Data*. <https://github.com/soodoku/dp-data/tree/22f46c19814202ea6c249109d7c2c3c76517ec9b>.
- Steele, Claude M. 1997. “A Threat in the Air: How Stereotypes Shape Intellectual Identity and Performance.” *American Psychologist* 52 (6): 613–29. <https://doi.org/10.1037/0003-066x.52.6.613>.
- Sturgis, Patrick, Caroline Roberts, and Nick Allum. 2005. “A Different Take on the Deliberative Poll: Information, Deliberation, and Attitude Constraint.” *Public Opinion Quarterly* 69 (1): 30–65. <https://doi.org/10.1093/poq/nfi005>.
- Theuwis, Marie-Isabel, Carolien Van Ham, and Kristof Jacobs. 2025. “A Meta-Analysis of the Effects of Democratic Innovations on Participants’ Attitudes, Behaviour and Capabilities.” *European Journal of Political Research* 64 (2): 960–84. <https://doi.org/10.1111/1475-6765.12722>.
- van Breukelen, Gerard J. P. 2006. “ANCOVA Versus Change from Baseline: More Power in Randomized Studies, More Bias in Nonrandomized Studies.” *Journal of Clinical Epidemiology* 59 (9): 920–25. <https://doi.org/10.1016/j.jclinepi.2006.02.007>.

A The Deliberative Polls

These 28 polls supply the attendee sample throughout the paper. Three also provide control comparisons, including Northern Ireland’s T3-only contrast. Tables A1 and A2 list polls in the same chronological order.

Table A1: Polls with respondent-level knowledge-item answers at both waves.

Poll	Year	Mode	Control
UK Crime	1994	In person	No
UK EU	1995	In person	No
Central Power & Light	1996	In person	No
National Issues Convention	1996	In person	No
Southwestern Electric Power	1996	In person	No
UK Monarchy	1996	In person	No
West Texas Utilities	1996	In person	No
UK General Election	1997	In person	No
UK Health	1998	In person	No
Australia Republic Referendum	1999	In person	No
Bulgarian National Crime Poll	2002	In person	No
New Haven, CT	2002	In person	No
By the People: National	2003	Online	No
National Issues Convention 2	2003	In person	No
By the People 2004 Online Primaries	2004	Online	No
By the People 2004 US General Election	2004	Online	No
By the People 2004 US Presidential Primaries	2004	Online	No
By the People: Health and Education	2005	Online	No
Zeguo Township	2005	In person	No
By the People 2007	2007	Online	No
Northern Ireland	2007	In person	Yes
Tomorrow’s Europe (EU)	2007	In person	No
San Mateo, CA	2008	In person	No
Europolis	2009	In person	No
Michigan	2009	In person	No
California What’s Next	2011	In person	No
America in One Room	2019	In person	Yes
America in One Room: Climate	2021	Online	Yes

B Further results

Table A2: Initial knowledge (percent correct), observed gains, and guessing-adjusted learning (percentage points) across the analyzed polls.

Poll	Initial (%)	Gain (pp)	Adjusted (pp)
UK Crime	52.2	9.5	15.0
UK EU	51.4	16.1	20.8
Central Power & Light	29.4	16.1	20.3
National Issues Convention	46.5	1.2	5.1
Southwestern Electric Power	43.5	20.3	30.1
UK Monarchy	64.8	14.6	21.7
West Texas Utilities	39.9	28.8	35.8
UK General Election	54.5	10.3	18.2
UK Health	65.8	7.9	11.0
Australia Republic Referendum	51.9	24.0	33.8
Bulgarian National Crime Poll	43.2	9.3	12.9
New Haven, CT	36.8	22.3	28.0
By the People: National	39.3	3.5	4.7
National Issues Convention 2	37.3	12.3	15.0
By the People 2004 Online Primaries	47.9	4.6	5.5
By the People 2004 US General Election	75.2	6.9	9.4
By the People 2004 US Presidential Primaries	51.5	11.8	14.8
By the People: Health and Education	35.2	5.9	7.2
Zeguo Township	31.3	11.2	14.1
By the People 2007	50.7	11.2	13.2
Northern Ireland	21.8	28.6	32.8
Tomorrow's Europe (EU)	36.6	16.6	22.3
San Mateo, CA	14.5	17.2	19.6
Europolis	19.7	16.6	19.4
Michigan	34.2	9.8	12.3
California What's Next	60.4	21.4	32.1
America in One Room	45.8	14.3	17.4
America in One Room: Climate	68.6	12.2	16.7

Table A3: Average learning across polls, with hierarchical-bootstrap confidence intervals.

Quantity	Weighting	Polls	Estimate	95% interval
Observed gain (pp)	Equal	28	13.7	11.4 to 16.2
Guessing-adjusted learning (pp)	Equal	28	18.2	15.7 to 21.4
Observed gain (pp)	Precision	28	12.3	10.2 to 14.6
Guessing-adjusted learning (pp)	Precision	28	15.5	13.4 to 18.2
Observed gain: in person (pp)	Equal	21	15.6	12.8 to 18.7
Observed gain: online (pp)	Equal	7	8.0	5.4 to 10.8

Observed gain is T2 minus T1; guessing-adjusted learning estimates the percentage who learned an item, averaged across questions. Equal weighting gives each poll the same weight. Precision weighting uses $w_d = 1/\widehat{\text{Var}}(\hat{\theta}_d)$, with sampling variances estimated by resampling discussion groups within each poll. The weighted average is $\sum_d w_d \hat{\theta}_d / \sum_d w_d$. These weights are held fixed when resampling polls and groups for its interval. The mode rows split the equal-poll average into in-person and online polls.

Table A4: How the baseline-adjusted attendee–control difference varies with initial knowledge and education.

Study	Comparison	Difference (pp)	95% interval	N
America in One Room 2019	Initial score +10 pp	-1.5	-2.5 to -0.8	1,367
America in One Room 2019	Bachelor’s degree	2.7	-1.6 to 7.2	1,367
America in One Room: Climate 2021	Initial score +10 pp	-2.5	-3.1 to -2.0	1,633
America in One Room: Climate 2021	Bachelor’s degree	-4.7	-7.9 to -1.6	1,633

Table A4 reports changes in the attendee–control difference, not changes in everyone’s knowledge. N is the regression sample size. Degree compares a bachelor’s degree with no degree. A negative entry means the attendee advantage is smaller for people with higher initial knowledge or a bachelor’s degree. For example, the America in One Room initial-knowledge estimate of -1.5 percentage points compares two people whose starting scores differ by ten percentage points. Each regression includes attendance, initial score, the listed characteristic, and its interaction with attendance. Centering initial scores at their sample mean does not change the reported interaction. All intervals use the hierarchical bootstrap. Selection into attendance and score ceilings limit causal interpretation.

B.1 Additional participant characteristics

Age, education, and gender are harmonized for all 28 polls. The attitude models require additional indices, which are harmonized for 21 polls.

Education. Education, qua years of schooling, is distinct from, and only moderately correlated with, political knowledge. Controlling for cognitive ability, with which it is more strongly correlated, its

effect on knowledge appears negligible (Luskin 1990). But absent a separate measure of cognitive ability, it can roughly proxy the latter. Naturally, we expect it to have a positive effect. There may, however, be some mutual substitution with initial knowledge. People who know enough already may not need to be as bright to learn a lot, and people who are bright enough may not need to know as much to do so. We measure education as comparably as possible across polls in three categories: below secondary completion, secondary/some college, and a degree or higher.

Gender and age. These demographic variables exhibit real-world political knowledge gaps, with men knowing more than women and older people more than younger ones (Delli Carpini and Keeter 1996). Age is measured in decades and gender as a dummy scored 1 for women.

Proportion of women. Whether from greater comfort, more widely shared cognitive or communication style, or lower stereotype threat (Steele 1997), women may learn more in small groups containing more women. We calculate the proportion excluding the participant herself.

Heterogeneity of opinion. In the real world, people often know congenial but not uncongenial facts—those commonly cited by those sharing their views but not those commonly cited by those holding opposing views. Part of deliberation’s effect on learning may consist in getting people to note and accept (if, in many cases, still argue around) uncongenial facts. This can be expected to occur mainly on site. Participants learn a great deal in anticipation of the event, but the learning at that stage is mainly of congenial facts (Hansen 2004). On site, the participants find it harder to avoid confronting the uncongenial ones. This may happen more, the more opposing points of view are represented and argued in the small group—the more attitudinally heterogeneous the group is. We gauge this as the generalized variance, the determinant of the covariance matrix, of the group’s policy attitudes (Anderson 2003).

Attitude extremity. Participants starting with more extreme views may have shuttered out more uncongenial facts, leaving them with more ground for deliberation to make up. We measure extremity as the mean, across attitude indices j , of $|A_{ij} - .5|$, where A_{ij} is the i th respondent’s score on the j th index.

Reading the briefing materials. One of the main information vectors is the briefing document sent to participants shortly after recruitment, normally well in advance of the on-site deliberations. It provides some background facts and lays out the major arguments for and against given policy options, which embed additional facts congenial to each argument. Respondent-linked reading reports are available for 12 polls. They describe reading before discussion, recalled in the post-deliberation questionnaire. We score their ordered categories from zero to one using the source-specific response scales and analyze them in a separate model. The reading model uses the demographic specification with poll fixed effects and a small-group random intercept.

B.2 Models with additional controls

Table A5: Change in proportion correct (T2 minus T1) with additional controls. Coefficients with 95 percent hierarchical-bootstrap confidence intervals.

	Additional controls	Reading reports
Intercept	.076 [-.024, .181]	.450 [.099, .565]
T1 knowledge	-.526 [-.644, -.394]	-.594 [-.974, -.287]
Secondary/some college	.118 [.058, .184]	.097 [-.027, .227]
BA or more	.194 [.132, .262]	.181 [.069, .302]
T1 x secondary/some college	-.040 [-.163, .075]	-.025 [-.333, .376]
T1 x BA or more	-.097 [-.227, .022]	-.053 [-.356, .339]
Age (decades)	-.003 [-.009, .004]	-.004 [-.014, .006]
Attitude extremity	-.078 [-.178, .035]	
Group size	.000 [-.003, .003]	-.002 [-.006, .002]
Groupmates' mean T1	.048 [-.125, .253]	-.004 [-.208, .169]
Opinion heterogeneity	.004 [-.241, .252]	
Female	-.020 [-.077, .041]	-.028 [-.106, .045]
Group share women	-.015 [-.095, .062]	-.038 [-.140, .052]
Female x group share women	.028 [-.090, .138]	.012 [-.129, .149]
Online	-.097 [-.148, -.041]	
Poll's mean initial score	.533 [.267, .764]	
Briefing reading		.063 [.032, .095]
Participants	5,728	3,608
Small groups	397	252
Polls	21	12
Marginal R-squared	0.269	0.355
Conditional R-squared	0.325	0.360

Notes. The model with additional controls includes poll and small-group random intercepts; the reading-reports model adds reading to the demographic specification, using poll fixed effects and small-group random intercepts. Attitude measures are omitted from that model. Adding groupmates' T1 correctness on questions the respondent missed to the model with additional controls gives -.049 [-.171, .077] among 5,728 participants. Respondents with no missed T1 items receive zero on this measure and a separate indicator (-.026 [-.115, .052]); none is removed for lack of an opportunity to learn. Marginal R^2 covers fixed effects; conditional R^2 includes random intercepts.

B.3 Matched-sample comparisons

Table A6 separates sample restrictions from added covariates. Column 1 repeats the baseline model in all polls. Column 2 restricts it to the polls with the additional attitude measures. Columns 3 and 4 use exactly the same participants: column 3 retains the baseline specification, and column 4 adds the controls shown in Table A5. Thus the final comparison holds sample composition fixed.

Table A6: Baseline coefficients under sample restrictions and additional controls. 95 percent hierarchical-bootstrap confidence intervals appear below coefficients.

	All polls	21 polls	Same cases	Added controls
Intercept	.194 [.128, .266]	.187 [.132, .262]	.185 [.134, .258]	.076 [-.024, .181]
T1 knowledge	-.490 [-.553, -.436]	-.532 [-.596, -.470]	-.532 [-.598, -.470]	-.526 [-.644, -.394]
Group size	-.001 [-.003, .001]	.000 [-.004, .002]	.000 [-.003, .002]	.000 [-.003, .003]
Groupmates' mean T1	.048 [-.086, .198]	.064 [-.115, .284]	.052 [-.120, .264]	.048 [-.125, .253]
Online	-.067 [-.109, -.017]	-.072 [-.124, -.015]	-.072 [-.123, -.013]	-.097 [-.148, -.041]
Poll's mean initial score	.386 [.183, .584]	.392 [.115, .630]	.402 [.138, .634]	.533 [.267, .764]
Participants	8,800	5,869	5,728	5,728
Small groups	592	397	397	397
Polls	28	21	21	21
Marginal R-squared	0.234	0.227	0.228	0.269
Conditional R-squared	0.298	0.293	0.291	0.325

The peer-knowledge coefficient is .064 [-.115, .284] in the 21-poll baseline model. Among the complete cases, it is .052 [-.120, .264] before adding controls and .048 [-.125, .253] afterward. The change from added controls is small relative to the uncertainty in these estimates.

The corresponding check for the demographic model uses its 8,671 participants. On these same cases, the baseline peer-knowledge coefficient is .039 [-.090, .174], compared with .030 [-.095, .156] after adding demographics.

B.4 Additional control comparisons

Regressing T2 scores on attendance and T1 scores answers a different question from comparing gains. The baseline-adjusted attendee-control differences are 14.2 percentage points in America in One Room (12.3 to 16.2), and 12.1 in the climate poll (10.4 to 13.7). These intervals use the same bootstrap as the main comparisons.

A year after the climate poll, the baseline-adjusted gap is 5.5 percentage points (3.5 to 7.4). Northern Ireland has only a follow-up control interview: 93 returning attendees scored 17.7 percentage points above 150 controls (12.3 to 23.5). This is a cross-sectional comparison without a control baseline. We do not apply the immediate-learning model's no-forgetting assumption to these later outcomes.

B.5 Poll-specific peer associations

The following estimates fit a separate slope in each poll, conditional on own initial knowledge. Their equal-poll average is .080 [-.094, .247].

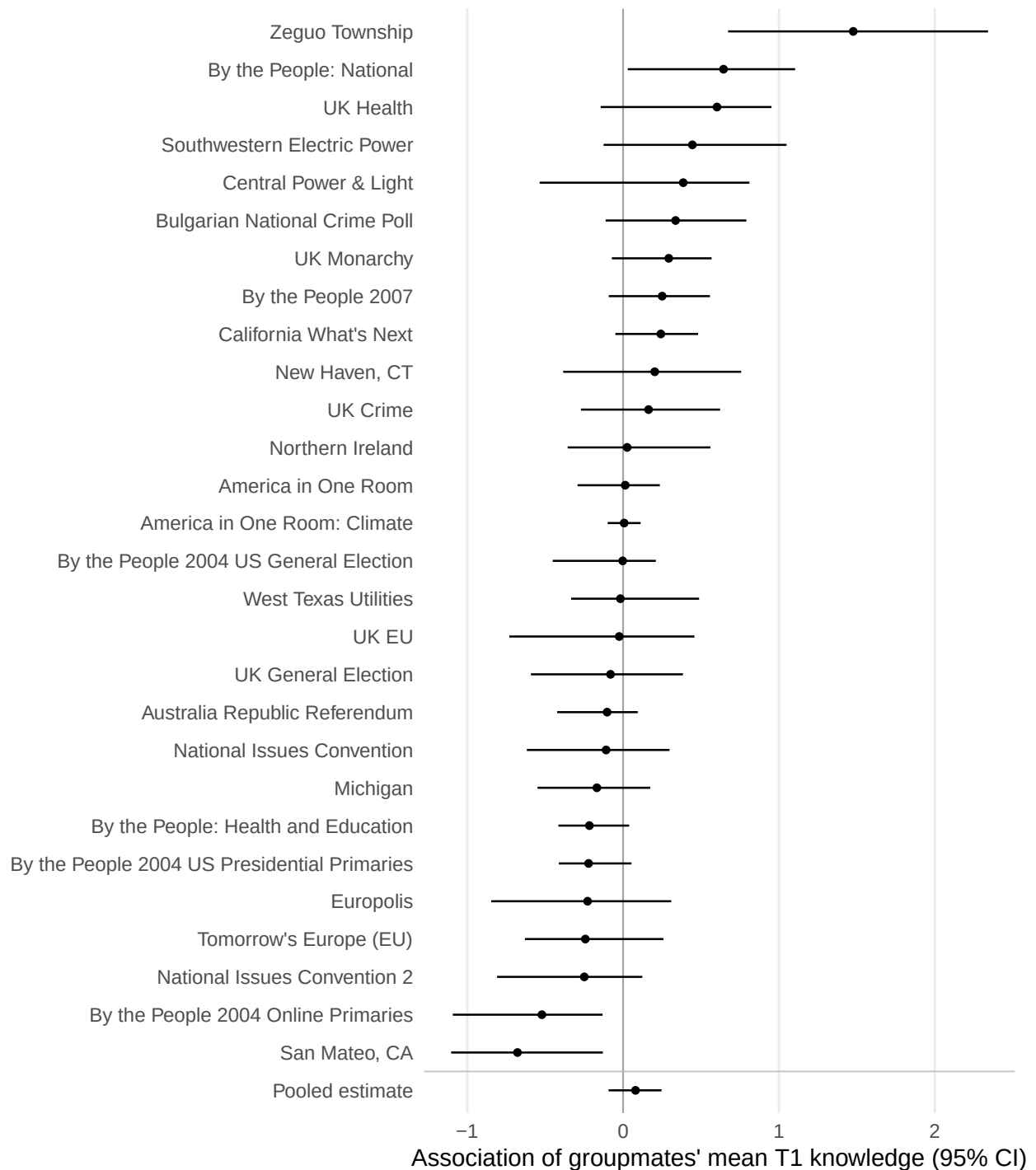


Figure A1: Association between the other group members' mean T1 knowledge and own knowledge gain, by poll, with 95% intervals. Each estimate controls for own T1 knowledge. Intervals use the hierarchical bootstrap. The pooled estimate weights polls equally and resamples polls within mode and then their groups. These are descriptive associations.

B.6 Baseline composition of discussion groups

We regress each participant’s baseline characteristic on its mean among the other group members, controlling for the mean among other members of the poll and poll fixed effects (Guryan et al. 2009). This is a descriptive sorting diagnostic; it does not reconstruct the original assignment mechanism. A randomization test would require the original assignment pools and restrictions, including any scheduling or language constraints. Balance on observed covariates alone does not establish random assignment.

Table A7: Baseline composition of discussion groups. Associations with 95 percent hierarchical-bootstrap intervals.

Characteristic	Association	Participants
Initial knowledge	.071 [-.026, .166]	8,800
Female	.007 [-.055, .066]	8,774
Age	-.010 [-.064, .041]	8,741
Degree or higher	.030 [-.038, .096]	8,709

C Knowledge items

The catalog lists 221 questions for the same 28 polls used in the learning analysis throughout the paper (Sood 2026). Wording is paraphrased where only source labels survive; capitalization and unambiguous abbreviations are normalized for readability. Original labels and scoring rules are preserved in the cited dataset. Choices are transcribed where recoverable; truncated or unavailable options are identified. Notes explain open-response coding.

C.1 UK Crime (1994)

- **Can British courts sentence a murderer to death?** (true/false). Choices: True | False. Scored correct: False [0].
- **Does Britain have the largest prison population in Western Europe?** (true/false). Choices: True | False. Scored correct: True [1].
- **Does Britain have more prisoners serving life sentences than the rest of the European Community combined?** (true/false). Choices: True | False. Scored correct: True [1].
- **Can a defendant be tried by jury in a local magistrates court?** (true/false). Choices: True | False. Scored correct: False [0].
- **Is the number of members of Parliament about 100?** (true/false). Choices: True | False. Scored correct: False [0].
- **Is the longest permitted interval between general elections four years?** (true/false). Choices: True | False. Scored correct: False [0].
- **Is the British electoral system based on proportional representation?** (true/false). Choices: True | False. Scored correct: False [0].

C.2 UK EU (1995)

- **Has the European Union recently expanded to 15 members?** (true/false). Choices: True | False. Scored correct: True [1].
- **Is Switzerland about to join the European Union?** (true/false). Choices: True | False. Scored correct: False [2].
- **Are British income tax rates decided in Brussels?** (true/false). Choices: True | False. Scored correct: False [2].
- **Are European Parliament elections held every five years?** (true/false). Choices: True | False. Scored correct: True [1].
- **Are the Liberal Democrats the least favorable to the European Union of the three main British parties?** (true/false). Choices: True | False. Scored correct: False [2].

C.3 Central Power & Light (1996)

- **Which energy source produces the greatest amount of CPL electricity?** (multiple choice). Choices: 1: Coal | 2: Wind | 3: Natural Gas | 4: Fuel Oil | 5: Nuclear | 6: Solar | 99: dk. Scored correct: Natural Gas [3].
- **Which customer group uses the most CPL electricity?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Industrial [3].
- **Which CPL customer group pays the highest electric rates?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Commerical [2].
- **Which electricity source produces the most air emissions?** (multiple choice). Choices: 1: Nuclear | 2: Coal | 3: Natural Gas | 99: dk. Scored correct: Coal [2].
- **Which agency sets electric rates in Texas?** (open coded). Scored correct: Puc [1]. Open answer coded into source categories; code 1 (Public Utility Commission) scored correct. Other codes scored incorrect.
- **How much profit do Texas electric utilities make on fuel?** (multiple choice). Choices: 0: 0% | 1: 10% | 2: 20% | 3: 30% | 99: dk. Scored correct: 0% [0].
- **How much of the electric bill pays for fuel?** (multiple choice). Choices: 1: less than 20% | 2: 21-30% | 3: 31-40% | 4: 41-50% | 5: More Than 50% | 99: dk. Scored correct: 21-30% [2].

C.4 National Issues Convention (1996)

- **About what percentage of American children are born outside marriage?** (open numeric). Scored correct: 25-40 inclusive [25-40 inclusive]. Numeric response scored correct when within the inclusive interval shown in the answer key; missing and out-of-range responses scored incorrect.
- **About what percentage of children live in single-parent households?** (open numeric). Scored correct: 1-10 inclusive [1-10 inclusive]. Numeric response scored correct when within

the inclusive interval shown in the answer key; missing and out-of-range responses scored incorrect.

- **What was the unemployment rate over the previous ten years?** (open numeric). Scored correct: 5-10 inclusive [5-10 inclusive]. Numeric response scored correct when within the inclusive interval shown in the answer key; missing and out-of-range responses scored incorrect.
- **Which listed purpose does the United States spend the most on?** (multiple choice). Choices: 1: The space program | 2: Foreign aid | 3: National defense | 4: Agriculture subsidies | 8: Do not know. Scored correct: National defense [3].
- **Which listed country does the United States trade with most?** (multiple choice). Choices: 1: Canada | 2: Japan | 3: Germany | 4: Mexico | 8: Do not know. Scored correct: Canada [1].
- **Had the United States sent ground troops to Iraq in the previous ten years?** (yes/no). Choices: Yes | No. Scored correct: Yes [1].
- **Had the United States sent ground troops to Somalia in the previous ten years?** (yes/no). Choices: Yes | No. Scored correct: Yes [1].
- **Had the United States sent ground troops to Rwanda in the previous ten years?** (yes/no). Choices: Yes | No. Scored correct: No [0].
- **Had the United States sent ground troops to Bosnia in the previous ten years?** (yes/no). Choices: Yes | No. Scored correct: Yes [1].
- **Where were Republicans on the liberal to conservative scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Slightly conservative | Conservative | Extremely conservative [5|6|7].
- **Where were Democrats on the liberal to conservative scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Extremely liberal | Liberal | Slightly liberal [1|2|3].

C.5 Southwestern Electric Power (1996)

- **Which energy source produces the greatest amount of SWEPCO electricity?** (multiple choice). Choices: 1: Coal | 2: Wind | 3: Natural Gas | 4: Fuel Oil | 5: Nuclear | 6: Solar | 99: dk. Scored correct: Coal [1].
- **Which customer group uses the most SWEPCO electricity?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Industrial [3].
- **Which SWEPCO customer group pays the highest electric rates?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Residential [1].
- **Which electricity source produces the most air emissions?** (multiple choice). Choices: 1: Nuclear | 2: Coal | 3: Natural Gas | 99: dk. Scored correct: Coal [2].

- **Which agency sets electric rates in Texas?** (open coded). Scored correct: Puc [1]. Open answer coded into source categories; code 1 (Public Utility Commission) scored correct. Other codes scored incorrect.

C.6 UK Monarchy (1996)

- **Is Princess Anne next in line to the throne?** (true/false). Choices: True | False. Scored correct: 2 false [2].
- **Is the prime minister, rather than the Queen, Britain's head of state?** (true/false). Choices: True | False. Scored correct: 2 false [2].
- **Is the Queen head of the Commonwealth?** (true/false). Choices: True | False. Scored correct: 1 true [1].
- **Is it the Queen's duty to decide the date of a general election?** (true/false). Choices: True | False. Scored correct: 2 false [2].
- **Does a prime minister take office only after being asked by the Queen?** (true/false). Choices: True | False. Scored correct: 1 true [1].
- **Is the Queen head of the Church of England?** (true/false). Choices: True | False. Scored correct: 1 true [1].
- **Is the Queen head of the Church of Scotland?** (true/false). Choices: True | False. Scored correct: 2 false [2].
- **Does Britain have an unwritten constitution?** (true/false). Choices: True | False. Scored correct: 1 true [1].
- **If a British monarch dies, who succeeds them?** (multiple choice). Choices: 1: 1 brother or siste | 2: 2 eldest brother | 3: 3 eldest sister | 4: 4 eldest child | 5: 5 eldest son | 6: 6 eldest daughter | 7: 7 none of these | 8: 8 can't choose | 9: 9 not answered. Scored correct: 5 eldest son [5].

C.7 West Texas Utilities (1996)

- **Which energy source produces the greatest amount of WTU electricity?** (multiple choice). Choices: 1: Coal | 2: Wind | 3: Natural Gas | 4: Fuel Oil | 5: Nuclear | 6: Solar | 99: dk. Scored correct: Natural Gas [3].
- **Which customer group uses the most WTU electricity?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Residential [1].
- **Which WTU customer group pays the highest electric rates?** (multiple choice). Choices: 1: Residential | 2: Commerical | 3: Industrial | 99: dk. Scored correct: Residential [1].
- **Which electricity source produces the most air emissions?** (multiple choice). Choices: 1: Nuclear | 2: Coal | 3: Natural Gas | 99: dk. Scored correct: Coal [2].
- **Which agency sets electric rates in Texas?** (open coded). Scored correct: Puc [1]. Open answer coded into source categories; code 1 (Public Utility Commission) scored correct. Other codes scored incorrect.

C.8 UK General Election (1997)

- **Had inflation been below five percent over the previous few years?** (true/false). Choices: True | False. Scored correct: True [1].
- **Did the Bank of England decide interest rates?** (true/false). Choices: True | False. Scored correct: False [2].
- **Was unemployment in the United Kingdom higher than in Germany?** (true/false). Choices: True | False. Scored correct: False [2].
- **Where was the Conservative Party on the income redistribution scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Gov Do Nothing to Make Inc. = | | [1|2|3].
- **Where was the Conservative Party on the tax and public services scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Gov Spend Less, Cut Taxes Lot | | [1|2|3].
- **Where was the Conservative Party on the minimum wage scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Yes Min. Wage | | [1|2|3].
- **Where was the Conservative Party on the European Union scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Keep Distance from E.U. | | [1|2|3].
- **Where was the Labour Party on the income redistribution scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Gov Try Hard Make Inc. = [5|6|7].
- **Where was the Labour Party on the tax and public services scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Gov Spend Lot, Tax lot [5|6|7].
- **Where was the Labour Party on the minimum wage scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Positions 5, 6, 7 [5|6|7].
- **Where was the Labour Party on the European Union scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Unite with E.U. [5|6|7].
- **Where were the Liberal Democrats on the income redistribution scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Gov Try Hard Make Inc. = [5|6|7].
- **Where were the Liberal Democrats on the tax and public services scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Gov Spend Lot, Tax lot [5|6|7].
- **Where were the Liberal Democrats on the minimum wage scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: Positions 5, 6, 7 [5|6|7].
- **Where were the Liberal Democrats on the European Union scale?** (placement scale). Choices: See source instrument for scale endpoints. Scored correct: | | Unite with E.U. [5|6|7].

C.9 UK Health (1998)

- **Was the standard National Health Service prescription charge ten pounds?** (true/false). Choices: True | False. Scored correct: false [0].
- **Was the proportion of older people in Britain increasing?** (true/false). Choices: True | False. Scored correct: true [1].
- **Had inflation-adjusted National Health Service spending doubled in the previous 20 years?** (true/false). Choices: True | False. Scored correct: true [1].
- **Did most people use private medical treatment instead of the National Health Service?** (true/false). Choices: True | False. Scored correct: false [0].
- **All British women can get free breast cancer screening on the NHS.** (true/false). Choices: True | False. Scored correct: false [0]. UKH-12: use exact universal availability wording; codebook opening caveat distinguishes literal false from anticipated true. Maintained false key is preserved; printed correctness counts conflict with the maintained SAV. No rekeying authorized.
- **Was cosmetic surgery never available through the National Health Service?** (true/false). Choices: True | False. Scored correct: false [0].

C.10 Australia Republic Referendum (1999)

- **How was the Governor-General appointed under the existing monarchy?** (multiple choice). Choices: 1: Chooses Governor-General | 2: No role | 3: Appointed on recommendation of Parliament | 4: Appointed on advice of prime minister (correct) | 96: None | 97: Don't know. Scored correct: Appointed on advice of prime minister (correct) [4].
- **Could the prime minister remove the proposed president?** (multiple choice). Choices: 1: Anytime without report to House | 2: Anytime with later House approval (correct) | 3: After fair trial in Parliament | 4: Cannot remove | 96: None | 97: Don't know. Scored correct: Anytime with later House approval (correct) [2].
- **Could the Governor-General dismiss the government?** (multiple choice). Choices: 1: Controls government | 2: Acts on Queen's instructions | 3: Performs only ceremonial duties | 4: Can decide to dismiss government (correct) | 96: None | 97: Don't know. Scored correct: Can decide to dismiss government (correct) [4].
- **Which existing office would the proposed president most resemble?** (multiple choice). Choices: 1: British prime minister | 2: Australian prime minister | 3: American president | 4: Governor-General (correct) | 96: None | 97: Don't know. Scored correct: Governor-General (correct) [4].
- **Which party was more concerned with social and welfare issues?** (multiple choice). Choices: 1: Liberal Party more | 2: Equally concerned | 3: Labor Party more (correct) | 96: None | 97: Not much impression/don't know. Scored correct: Labor Party more (correct) [3].
- **Which party was closer to business?** (multiple choice). Choices: 1: Liberal Party more (correct) | 2: Equally close | 3: Labor Party more | 96: None | 97: Not much impression/don't know. Scored correct: Liberal Party more (correct) [1].

- **What public office did Aden Ridgeway hold?** (multiple choice). Choices: 1: Leader of federal opposition | 2: Leader of Democrats | 3: Justice of the High Court | 4: Aboriginal senator in Parliament (correct) | 5: None of these | 6: Don't know | 97: Don't know. Scored correct: Aboriginal senator in Parliament (correct) [4].
- **What public office did Jennie George hold?** (multiple choice). Choices: 1: Secretary of Teachers Federation | 2: Labor member of federal Parliament | 3: Leader of workers' union | 4: President of actu (correct) | 5: None of these | 97: Don't know. Scored correct: President of actu (correct) [4].
- **Would the Australian flag change under the proposed republic?** (multiple choice). Choices: 1: No | 2: Yes. Scored correct: No [1].
- **Would the national anthem change under the proposed republic?** (multiple choice). Choices: 1: No | 2: Yes. Scored correct: No [1].
- **Would use of the word Royal in military service names change under the proposed republic?** (multiple choice). Choices: 1: No | 2: Yes. Scored correct: Yes [2].
- **Would Australian participation in the Commonwealth Games change under the proposed republic?** (multiple choice). Choices: 1: No | 2: Yes. Scored correct: No [1].

C.11 Bulgarian National Crime Poll (2002)

- **Could police detain a citizen for questioning at any hour without allowing refusal?** (true/false). Choices: True | False. Scored correct: False [2].
- **Could a court detain a suspect indefinitely while proving guilt?** (true/false). Choices: True | False. Scored correct: False [2].
- **Could prosecutors detain a suspect until a case was resolved?** (true/false). Choices: True | False. Scored correct: False [2].
- **Was the chief prosecutor accountable to parliament?** (true/false). Choices: True | False. Scored correct: False [2].
- **Were police forbidden to use violence against a detainee even if the person was guilty?** (true/false). Choices: True | False. Scored correct: True [1].
- **Could police detain a person for 72 hours?** (true/false). Choices: True | False. Scored correct: False [2].
- **Was a confession alone insufficient to establish guilt?** (true/false). Choices: True | False. Scored correct: True [1].

C.12 New Haven, CT (2002)

- **Which population estimate is closest to that of the greater New Haven region?** (multiple choice). Choices: 1: 250,000 | 2: 350,000 | 3: 550,000 | 4: 750,000. Scored correct: 550,000 [3].
- **Was New Haven regional job growth in the 1990s greater than, similar to, or less than the rest of the United States?** (multiple choice). Choices: 1: More | 2: About the same | 3: Less. Scored correct: About the same [2].

- **What is the main source of revenue for most town governments in greater New Haven?** (multiple choice). Choices: 1: Sales taxes | 2: Property taxes | 3: State subsidies | 4: Federal subsidies. Scored correct: Property taxes [2].
- **Did the population of New Haven city increase, stay the same, or decrease in the 1990s?** (multiple choice). Choices: 1: Increased | 2: Stayed the same | 3: Decreased. Scored correct: Decreased [3].
- **Does Connecticut law allow communities to share property tax revenues?** (multiple choice). Choices: 1: Yes | 2: No. Scored correct: Yes [1].
- **Do communities with the most valuable property tend to have the lowest, average, or highest property tax rates?** (multiple choice). Choices: 1: Lowest | 2: Average | 3: Highest. Scored correct: Lowest [1].
- **How does the Federal Aviation Authority classify Tweed New Haven Airport?** (multiple choice). Choices: 1: Major hub | 2: Medium hub | 3: Minor hub | 4: Non-hub. Scored correct: Non-hub [4].
- **Would maintaining Tweed New Haven Airport at its current level of service require significant investment?** (multiple choice). Choices: 1: Yes | 2: No. Scored correct: Yes [1].

C.13 By the People: National (2003)

- **Where did the Democratic Party stand on a 0–10 scale from doing nothing about global warming to acting now?** (placement scale). Choices: 0–10 scale: 0 = do nothing | 5 = midway | 10 = act now. Scored correct: Above midpoint [6|7|8|9|10].
- **Where did the Republican Party stand on a 0–10 scale from doing nothing about global warming to acting now?** (placement scale). Choices: 0–10 scale: 0 = do nothing | 5 = midway | 10 = act now. Scored correct: Below midpoint [0|1|2|3|4].
- **Did President Bush want to increase, decrease, or maintain foreign aid?** (multiple choice). Choices: 1: increase | 2: decrease | 3: keep it the same | 4: haven't heard anything about this. Scored correct: increase [1].
- **Did President Bush support recent international agreements to control greenhouse gases?** (multiple choice). Choices: 1: supports | 2: opposes | 3: haven't heard anything about this. Scored correct: opposes [2].
- **How much of each \$100 in the federal budget went to military spending?** (multiple choice). Choices: 1: 1 dollar, | 2: 5 dollars, | 3: 10 dollars, | 4: 20 dollars, | 5: 30 dollars or more, or | 6: don't you have much idea about that?. Scored correct: 20 dollars, [4].
- **How much of each \$100 in the federal budget went to foreign aid?** (multiple choice). Choices: 1: 1 dollar, | 2: 5 dollars, | 3: 10 dollars, | 4: 20 dollars, | 5: 30 dollars or more, or | 6: don't you have much idea about that?. Scored correct: 1 dollar, [1].
- **How much of each \$100 of United States output was sold abroad?** (multiple choice). Choices: 1: 4 dollars worth, | 2: 8 dollars, | 3: 12 dollars, | 4: 24 dollars or more, or | 5: don't you have much idea about that?. Scored correct: 12 dollars, [3].

- **About how many adults per 100 in the most affected African countries had HIV/AIDS?** (multiple choice). Choices: 1: fewer than 5 out of 100, | 2: about 10 of 100, | 3: about 20 of 100, | 4: 30 or more out of 100, or | 5: don't you have much idea about that?. Scored correct: 30 or more out of 100, or [4].
- **Did the United States have a veto over World Trade Organization decisions?** (multiple choice). Choices: 1: true | 2: false. Scored correct: false [2].
- **Did the United States have a veto on the United Nations Security Council?** (multiple choice). Choices: 1: true | 2: false. Scored correct: true [1].
- **What was the main cause of global warming?** (multiple choice). Choices: 1: it is caused mostly by human activities, like driving cars a | 2: it is caused mostly by natural changes in the climate, | 3: it is not occurring at all, or | 4: haven't you thought much about this?. Scored correct: it is caused mostly by human activities, like driving cars a [1].

C.14 National Issues Convention 2 (2003)

- **Where did the Democratic Party stand on the global warming action scale?** (placement scale). Choices: 98: don't know. Scored correct: Above midpoint [6|7|8|9|10].
- **Where did the Republican Party stand on the global warming action scale?** (placement scale). Choices: 98: don't know. Scored correct: Below midpoint [0|1|2|3|4].
- **Did President Bush want to increase, decrease, or maintain foreign aid?** (multiple choice). Choices: 1: increase | 2: decrease | 3: keep it the same | 4: haven't heard anything about this | 8: don't know. Scored correct: increase [1].
- **Did President Bush support recent international agreements to control greenhouse gases?** (multiple choice). Choices: 1: supports | 3: opposes | 5: haven't heard anything about this | 8: don't know. Scored correct: opposes [3].
- **How much of each \$100 in the federal budget went to military spending?** (multiple choice). Choices: 1: 1 dollar | 2: 5 dollars | 3: 10 dollars | 4: 20 dollars | 5: 30+ dollars | 6: not much idea | 7: vol: other amount (specify) | 8: don't know. Scored correct: 20 dollars [4].
- **How much of each \$100 in the federal budget went to foreign aid?** (multiple choice). Choices: 1: 1 dollar | 2: 5 dollars | 3: 10 dollars | 4: 20 dollars | 5: 30+ dollars | 6: not much idea | 7: vol: other amount (specify) | 8: don't know. Scored correct: 1 dollar [1].
- **How much of each \$100 of United States output was sold abroad?** (multiple choice). Choices: 1: 4 dollars | 2: 8 dollars | 3: 12 dollars | 4: 24+ dollars | 5: not much idea | 7: vol: other amount (specify) | 8: don't know. Scored correct: 12 dollars [3].
- **About how many adults per 100 in the most affected African countries had HIV/AIDS?** (multiple choice). Choices: 1: fewer than 5 out of 100 | 2: about 10 of 100 | 3: about 20 of 100 | 4: 30+ out of 100 | 5: not much idea | 7: vol: other amount (specify) | 8: don't know. Scored correct: 30+ out of 100 [4].
- **Did the United States have a veto over World Trade Organization decisions?** (multiple choice). Choices: 1: true | 5: false | 8: don't know. Scored correct: false [5].

- **Did the United States have a veto on the United Nations Security Council?** (multiple choice). Choices: 1: true | 5: false | 8: don't know. Scored correct: true [1].
- **What was the main cause of global warming?** (multiple choice). Choices: 1: human activities | 2: natural changes | 3: not occurring at all | 4: haven't thought much about this | 8: don't know. Scored correct: human activities [1].

C.15 By the People 2004 Online Primaries (2004)

- **What did President Bush's tax cut do to taxes on investment income?** (multiple choice). Choices: 1: lower it | 2: keep it the same | 3: raise it | 4: don't know. Scored correct: lower it [1].
- **What was Wesley Clark's most recent government position?** (multiple choice). Choices: 1: senator from arkansas | 2: secretary of the air force | 3: supreme allied commander of nato | 4: national security advisor to the president | 5: don't know. Scored correct: supreme allied commander of nato [3].
- **Why did President Bush end tariffs on imported steel?** (multiple choice). Choices: 1: the steel industry and steelworkers unions objected | 2: he wanted to avoid increased tariffs by the european union o | 3: domestic steel production is insufficient to meet demand, th | 4: he has never supported tariffs | 5: don't know. Scored correct: he wanted to avoid increased tariffs by the european union o [2].
- **Which Democratic presidential candidate had served as House majority leader?** (multiple choice). Choices: 1: john edwards | 2: howard dean | 3: joe lieberman | 4: richard gephardt | 5: don't know. Scored correct: richard gephardt [4].
- **How had the unemployment rate changed since President Bush took office?** (multiple choice). Choices: 1: decreased | 2: stayed about the same | 3: increased | 4: don't know. Scored correct: increased [3].
- **How many Americans had been killed in action in Iraq?** (multiple choice). Choices: 1: fewer than a hundred | 2: several hundred | 3: several thousand | 4: more than twenty thousand | 5: don't know. Scored correct: several hundred [2].
- **What was the Bush administration's position on a Free Trade Area of the Americas?** (multiple choice). Choices: 1: support it | 2: neither support nor oppose it | 3: oppose it | 4: don't know. Scored correct: support it [1].

C.16 By the People 2004 US General Election (2004)

- **How did John Kerry vote on the Senate resolution authorizing the Iraq war?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: vote for it | 2: vote against it | 3: not vote on it | 4: blank | 5: don't know. Scored correct: vote for it [1].
- **Which country harbored the most Al Qaeda and Taliban fighters?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: india | 2: pakistan | 3: sri lanka | 4: indonesia | 5: blank | 6: don't know. Scored correct: pakistan [2].

- **Was Iraq directly involved in the September 11 attacks?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: true | 2: false | 3: blank | 4: don't know. Scored correct: false [2].
- **Had large quantities of weapons of mass destruction been found in Iraq?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: true | 2: false | 3: blank | 4: don't know. Scored correct: false [2].
- **Did prescription drugs cost more in Canada than in the United States?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: true | 2: false | 3: blank | 4: don't know. Scored correct: false [2].
- **What was George W. Bush's role during the Vietnam War?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: he was drafted but never went to vietnam | 2: he was a decorated officer serving in vietnam | 3: he was ineligible to serve in the military because of a medi | 4: he served in the texas air national guard | 5: blank | 6: don't know. Scored correct: he served in the texas air national guard [4].
- **What was John Kerry's role during the Vietnam War?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: he was drafted but never went to vietnam | 2: he was a decorated officer serving in vietnam | 3: he was ineligible to serve in the military because of a medi | 4: he served in the massachusetts air national guard | 5: blank | 6: don't know. Scored correct: he was a decorated officer serving in vietnam [2].
- **Which country was a major destination for offshore white-collar jobs?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: south africa | 2: japan | 3: brazil | 4: india | 5: blank | 6: don't know. Scored correct: india [4].
- **About how many Americans had been killed in Iraq since the war began?** (multiple choice). Choices: -4: case in not in wave of data collection | -3: case in wave, no survey on file | 1: 100 | 2: 500 | 3: 1,000 | 4: 10,000 | 5: blank | 6: don't know. Scored correct: 1,000 [3].

C.17 By the People 2004 US Presidential Primaries (2004)

- **What did President Bush's tax cut do to taxes on investment income?** (multiple choice). Choices: 1: lower it | 2: keep it the same | 3: raise it | 4: don't know. Scored correct: lower it [1].
- **What was Wesley Clark's most recent government position?** (multiple choice). Choices: 1: senator from arkansas | 2: secretary of the air force | 3: supreme allied commander of nato | 4: national security advisor to the president | 5: don't know. Scored correct: supreme allied commander of nato [3].
- **Why did President Bush end tariffs on imported steel?** (multiple choice). Choices: 1: the steel industry and steelworkers unions objected | 2: he wanted to avoid increased tariffs by the european union o | 3: domestic steel production is insufficient to meet demand, th | 4: he has never supported tariffs | 5: don't know. Scored correct: he wanted to avoid increased tariffs by the european union o [2].

- **Which Democratic presidential candidate had served as House majority leader?** (multiple choice). Choices: 1: john edwards | 2: howard dean | 3: joe lieberman | 4: richard gephardt | 5: don't know. Scored correct: richard gephardt [4].
- **How had the unemployment rate changed since President Bush took office?** (multiple choice). Choices: 1: decreased | 2: stayed about the same | 3: increased | 4: don't know. Scored correct: increased [3].
- **How many Americans had been killed in action in Iraq?** (multiple choice). Choices: 1: fewer than a hundred | 2: several hundred | 3: several thousand | 4: more than twenty thousand | 5: don't know. Scored correct: several hundred [2].
- **What was the Bush administration's position on a Free Trade Area of the Americas?** (multiple choice). Choices: 1: support it | 2: neither support nor oppose it | 3: oppose it | 4: don't know. Scored correct: support it [1].

C.18 By the People: Health and Education (2005)

- **Where did the United States rank among wealthy countries in student mathematics skills?** (multiple choice). Choices: 1: bottom 3 | 2: bottom 10 | 3: top 10 | 4: top 3 | 7: dk. Scored correct: bottom 10 [2].
- **Had it become harder or easier for foreigners to get United States work permits?** (multiple choice). Choices: 1: harder | 2: same | 3: easier | 7: dk. Scored correct: harder [1].
- **Has the gap between minority students and white students in math and reading tests at the elementary school level been ...?** (multiple choice). Choices: 1: increasing the last few years | 2: staying about the same | 3: decreasing the last few years | 7: Couldn't say. Scored correct: decreasing the last few years [3]. Approved BTPHE-04: both questionnaires Q17 p.4 match education briefing PDF p.2; code 3 replaces inherited code 1 at both waves and calibration; original source responses unchanged.
- **About how many Americans lacked health insurance?** (multiple choice). Choices: 1: 1 million | 2: 25 million | 3: 45 million | 4: 65 million | 5: 85 million | 7: dk. Scored correct: 45 million [3].
- **How much was the population age 65 and older expected to grow over the next 50 years?** (multiple choice). Choices: 1: stay about the same | 2: increase by about half | 3: double | 4: triple | 7: dk. Scored correct: double [3].
- **What share of uninsured families had low incomes?** (multiple choice). Choices: 1: a quarter | 2: fifty per cent | 3: two thirds | 4: ninety per cent | 7: dk. Scored correct: two thirds [3].

C.19 Zeguo Township (2005)

- **By how much did Zeguo Township revenue increase from 2003 to 2004?** (multiple choice). Choices: 1: 10.2% | 2: 20.1% | 3: 33.7% | 4: No increase. Scored correct: 33.7% [3].
- **How large is the floating population in Zeguo Township?** (multiple choice). Choices: 1: 300,000 | 2: 50,000 | 3: 120,000 | 4: 200,000. Scored correct: 120,000 [3].

- **Which listed product is not a major product of Zeguo Township?** (multiple choice). Choices: 1: Water pumps | 2: Shoes | 3: Plastic products | 4: Air compressors. Scored correct: Plastic products [3].
- **How many parks does Zeguo Township currently have?** (multiple choice). Choices: 1: Zero | 2: One | 3: Two | 4: Five | 5: Seven. Scored correct: Two parks [3].

C.20 By the People 2007 (2007)

- **About how many Americans were prevented from voting because of criminal convictions?** (multiple choice). Choices: 1: 50,000 | 2: 500,000 | 3: 5,000,000 | 4: 15,000,000 | 99: Couldn't say. Scored correct: 5,000,000 [3].
- **What was the purpose of gerrymandering?** (multiple choice). Choices: 1: Increase electoral competition | 2: Ensure a majority for one party over another | 3: Include independents in the primary process | 4: Make them much smaller | 99: Couldn't say. Scored correct: Ensure a majority for one party over another [2].
- **What share of eligible voters voted in the specified presidential election?** (multiple choice). Choices: 1: 10% | 2: 30% | 3: 50% | 4: 70% | 99: Couldn't say. Scored correct: 50% [3].
- **Which listed country had compulsory voting?** (multiple choice). Choices: 1: Ireland | 2: United Kingdom | 3: Australia | 4: South Korea | 99: Couldn't say. Scored correct: Australia [3].
- **Who had to register with Selective Service in the United States?** (multiple choice). Choices: 1: Nobody | 2: Men between 18 and 25 years old | 3: Both men and women between 18 and 25 years old | 4: Men and women who receive federal grants for education | 99: Couldn't say. Scored correct: Men between 18 and 25 years old [2].
- **What vote total must a United States presidential candidate receive to win?** (multiple choice). Choices: 1: Gets more votes than any other candidate | 2: Gets a majority of all the votes cast | 3: Gets more Electoral College votes than any other candidate | 4: Gets a majority of Electoral College votes | 99: Couldn't say. Scored correct: Gets a majority of Electoral College votes [4].
- **How often are congressional districts redrawn?** (multiple choice). Choices: 1: Every 5 years | 2: Every 10 years | 3: Every 15 years | 4: Every 20 years | 99: Couldn't say. Scored correct: Every 10 years [2].
- **Which two states held the earliest presidential primary events?** (multiple choice). Choices: 1: Nevada and South Carolina | 2: Iowa and New Hampshire | 3: Indiana and New Mexico | 4: Wyoming and Delaware | 5: Michigan and Florida | 99: Couldn't say. Scored correct: Iowa and New Hampshire [2].

C.21 Northern Ireland (2007)

- **What share of Northern Ireland schools had mostly Protestant or mostly Catholic pupils?** (multiple choice). Choices: 1: more than 50% | 2: 40-50% | 3: 20-30% | 4: 5-10% | 5: less than 1% | 9: don't know. Scored correct: 5-10% [4]. Archived variable/value labels are

truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.

- **By about what percentage had the number of children entering Omagh schools changed?** (multiple choice). Choices: 1: increased by 20% | 2: increased by 10% | 3: stayed about the same | 4: decreased by 10% | 5: decreased by 20% | 9: don't know. Scored correct: decreased by 10% [4]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.
- **What did the new entitlement framework require for the school curriculum?** (multiple choice). Choices: 1: every school provides all 14-year-olds with a choice of at 1 | 2: every child has the right to attend any school his or her pa | 3: every child be provided a free school meal every school day | 4: every child receives free textbooks | 9: don't know. Scored correct: every school provides all 14-year-olds with a choice of at 1... [1]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.
- **What did the new entitlement framework require for access to courses?** (multiple choice). Choices: 1: every child receives tuition in the language of his or her p | 2: every child receives free transportation to and from school | 3: every denominational group has the right to run its own scho | 4: one-third of all the subjects offered must be applied | 9: don't know. Scored correct: one-third of all the subjects offered must be applied... [4]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.
- **What did pupils in Northern Ireland typically do after compulsory schooling?** (multiple choice). Choices: 1: about one-quarter go directly into employment | 2: about one-quarter leave school to be unemployed | 3: about three-quarters of grammar school pupils attend univers | 4: about three-quarters of secondary school students attend fur | 9: don't know. Scored correct: about three-quarters of grammar school pupils attend univers... [3]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.
- **Which statement about Northern Ireland school funding was correct?** (multiple choice). Choices: 1: schools receive more funding for older pupils | 2: schools receive the same funding for all pupils, regardless | 3: schools receive more funding for younger pupils | 9: don't know. Scored correct: schools receive more funding for older pupils [1]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.
- **Which statement about the employing authority for Northern Ireland schools was correct?** (multiple choice). Choices: 1: the official employer for all teachers is the education and | 2: the official employer for all teachers is the department of | 3: the official employer for all teachers in catholic schools i | 4: the official employer for all teachers in voluntary grammar | 9: don't know. Scored correct: the official employer for all teachers in voluntary grammar... [4]. Archived variable/value labels are truncated; the listed substantive choices are partial transcriptions. The key is the archived numeric code.

C.22 Tomorrow's Europe (EU) (2007)

- **Which listed country was an official candidate to join the European Union?** (multiple choice). Choices: 1: Romania | 2: Montenegro | 3: Croatia | 4: Morocco | 5: Or couldn't you say about that ? | 6: Ref | 7: Dk. Scored correct: Croatia [3].
- **How were members of the European Parliament chosen?** (multiple choice). Choices: 1: Directly elected by the citizens of their country | 2: Elected for three year terms | 3: Elected by the parliament of their country | 4: Appointed by their national head of government | 5: Or couldn't you say about that ? | 6: Ref | 7: Dk. Scored correct: Directly elected by the citizens of their country [1].
- **Which European Union institutions adopted employment laws?** (multiple choice). Choices: 1: The European Commission, in some cases with the European Parliament | 2: The Council of Ministers, in some cases with the European Parliament | 3: The European Parliament by itself | 4: The European commission by itself | 5: Or couldn't you say about that ? | 6: Ref | 7: Dk. Scored correct: The Council of Ministers, in some cases with the European Parliament [2].
- **What was the European Union's role in unemployment benefits?** (multiple choice). Choices: 1: Finance the member states unemployment benefit systems | 2: Decide the level and length of unemployment benefits in the member states | 3: Require that the member states merge their unemployment benefit systems by 2010 | 4: Guarantee that all EU citizens have access to unemployment benefits where they live | 5: Or couldn't say about that ? | 6: Ref | 7: Dk. Scored correct: Guarantee that all EU citizens have access to unemployment benefits where they live [4].
- **To what was roughly one-third of the European Union budget devoted?** (multiple choice). Choices: 1: Helping the EU's less prosperous regions | 2: Subsidizing the EU's fishing industry | 3: Financing diplomatic missions abroad | 4: Maintaining the EU's administration and bureaucracy | 5: Or couldn't you say about that ? | 6: Ref | 7: Dk. Scored correct: Helping the EU's less prosperous regions [1].
- **How was the share of European Union adults age 65 or older expected to change by 2050?** (multiple choice). Choices: 1: About one quarter of what it is now | 2: About half of what it is now | 3: About the same as what it is now | 4: About twice what it is now | 5: About four times what it is now | 6: Or couldn't you say about that ? | 7: Ref | 8: Dk. Scored correct: About twice what it is now [4].
- **How much foreign aid did the European Union and its members give relative to the United States?** (multiple choice). Choices: 1: The EU and its member states give roughly four times as much | 2: The EU and its member states give roughly twice as much | 3: The EU and its member states and the US give about the same amount | 4: The US gives roughly twice as much | 5: The US gives roughly four times as much | 6: Or couldn't you say about that ? | 7: Ref | 8: Dk. Scored correct: The EU and its member states give roughly twice as much [2].
- **About what share of European Union member governments' spending went to foreign aid?** (multiple choice). Choices: 1: About 1% | 2: About 5% | 3: About 9% | 4: About 13% | 5: About 17% | 6: Or couldn't you say about that ? | 7: Ref | 8: Dk. Scored correct: About 1% [1].

- **Which listed country did not possess nuclear weapons?** (multiple choice). Choices: 1: Pakistan | 2: India | 3: North Korea | 4: Japan | 5: Or couldn't you say about that ? | 6: Ref | 7: Dk. Scored correct: Japan [4].
- **Where was Nicolas Sarkozy on the left to right political scale?** (placement scale). Choices: 1: 0 | 2: 1 | 3: 2 | 4: 3 | 5: 4 | 6: 5 | 7: 6 | 8: 7 | 9: 8 | 10: 9 | 11: 10 | 12: Ref | 13: Dk. Scored correct: 6 | 7 | 8 | 9 | 10 [7|8|9|10|11].
- **Where was Gordon Brown on the left to right political scale?** (placement scale). Choices: 1: 0 | 2: 1 | 3: 2 | 4: 3 | 5: 4 | 6: 5 | 7: 6 | 8: 7 | 9: 8 | 10: 9 | 11: 10 | 12: Ref | 13: Dk. Scored correct: 0 | 1 | 2 | 3 | 4 [1|2|3|4|5].

C.23 San Mateo, CA (2008)

- **What was the yearly gap between new homes needed and homes built in San Mateo County?** (multiple choice). Choices: 1: about 200 | 2: about 500 | 3: about 1,000 | 4: about 5,000 | 5: about 8,000 | 6: Or couldn't you say about that?. Scored correct: about 1,000 [3].
- **What was the median single-family home price in San Mateo County in 2007?** (multiple choice). Choices: 1: 550,000 | 2: 680,000 | 3: 750,000 | 4: 830,000 | 5: 950,000 | 6: Or couldn't you say about that?. Scored correct: 950,000 [5]. Source labels say \$950,000 for code 5; the March 12 pre-questionnaire says \$940,000. The administered version is not established; the numeric key remains 5.
- **What share of county households could afford a median-priced home in 2006?** (multiple choice). Choices: 1: 12% | 2: 23% | 3: 37% | 4: 43% | 5: 55% | 6: Or couldn't you say about that?. Scored correct: 12% [1].
- **What was the average rent for a two-bedroom apartment in 2007?** (multiple choice). Choices: 1: \$1,100 | 2: \$1,300 | 3: \$1,700 | 4: \$2,200 | 5: \$3,000 | 6: Or couldn't you say about that?. Scored correct: \$1,700 [3].
- **How much was county housing demand projected to grow from 2005 to 2025?** (multiple choice). Choices: 1: 20,000 | 2: 40,000 | 3: 60,000 | 4: 70,000 | 5: 90,000 | 6: Or couldn't you say about that?. Scored correct: 70,000 [4].
- **According to the same Housing Needs Study, among the new households that are expected to form in the County between 2005 and 2025, about what percentage, combined, would be households with low, very low and extremely low incomes?** (multiple choice). Choices: 1: Less than one-third | 2: Less than one-half | 3: About half | 4: More than one-half | 5: More than two-thirds | 6: Or couldn't you say about that?. Scored correct: About half [3]. SM-05 catalog wording repair: includes all three income groups named in Q24; correct code 3 and all scores unchanged. The retained onsite post questionnaire contains no knowledge battery, so departure wording is not independently established.
- **How many new homes were projected to be built at the current development rate?** (multiple choice). Choices: 1: 25,000 | 2: 50,000 | 3: 75,000 | 4: 100,000 | 5: Or couldn't you say about that?. Scored correct: 25,000 [1].
- **What percentage of San Mateo County's land is agricultural use, watershed, open space, wetlands, or parks?** (multiple choice). Choices: 1: less than 25% | 2: 33% | 3: 58%

| 4: 67% | 5: more than 75% | 6: Or couldn't you say about that?. Scored correct: more than 75% [5]. Question text is from the March 12 pre-questionnaire; answer codes and labels follow the source dictionary. The briefing chart groups agriculture/rangeland (40%) and open space (33%) separately and does not by itself establish the keyed threshold.

C.24 Europol (2009)

- **V1Q43. Is the main decision-making body of the European Union the...?** (multiple choice). Choices: 1: European Commission | 2: Council of Ministers | 3: European Parliament | 4: European Court of Auditors | 999: You couldn't say. Scored correct: Council of Ministers [2].
- **V1Q44. Only one of the following statements about the European Parliament is false. Which one is it?** (multiple choice). Choices: 1: It passes all EU laws | 2: It can dismiss the European Commission | 3: It can reject the budget proposed by the Council of Minister | 4: It is involved in decisions about the admission of new Membe | 999: You couldn't say. Scored correct: It passes all EU laws [1].
- **V1Q46. Which of the following is true of Blue card workers?** (multiple choice). Choices: 1: They can work anywhere in the EU | 2: They must have a university education | 3: They cannot bring family members to join them any faster tha | 4: They are subject to the Returns Directive | 999: You couldn't say. Scored correct: They must have a university education [2].
- **V1Q47. Which of the following is true about the ways in which immigration policy is currently made?** (multiple choice). Choices: 1: The EU sets the basic rules about entry and residency requir | 2: The EU decides how many immigrants can be admitted to each c | 3: Work permits for immigrants must be approved by the EU | 4: The EU plays no role in immigration policy. | 999: You couldn't say. Scored correct: The EU sets the basic rules about entry and residency requir [1].
- **V1Q49. The percentage of the EU's total energy consumption that comes from fossil fuels (coal, gas or oil) is about...?** (multiple choice). Choices: 1: 50% | 2: 60% | 3: 70% | 4: 80% | 999: You couldn't say. Scored correct: 80% [4].
- **V1Q50. Which of the following produces the most greenhouse gases?** (multiple choice). Choices: 1: China | 2: The European Union | 3: The United States | 4: India | 999: You couldn't say. Scored correct: China [1].

C.25 Michigan (2009)

- **Which party held the majority in the Michigan State Senate?** (open text). Scored correct: Republican Party [2]. Open party name at departure was lowercased and stripped of punctuation; party-name variants listed in the cited dataset were accepted. The baseline source stores a coded party category, shown in the answer key.
- **Which party held the majority in the Michigan State House of Representatives?** (open text). Scored correct: Democratic Party [1]. Open party name at departure was lowercased and stripped of punctuation; party-name variants listed in the cited dataset were accepted. The baseline source stores a coded party category, shown in the answer key.

- **Which state had an unemployment rate similar to Michigan?** (multiple choice). Choices: 1: Oregon | 2: West Virginia | 3: Ohio | 4: Louisiana. Scored correct: Oregon [1].
- **Under which condition could a person remain eligible for the Family Independence Program after 48 months?** (multiple choice). Choices: 1: Live in a county with high unemployment | 2: Fired from a job while on the Family Self-Sufficiency Plan | 3: Non-citizen on the Refugee Assistance Program | 4: Have a small child in the ChildFirst program. Scored correct: Live in a county with high unemployment [1].
- **About what percentage of African American children in Michigan lived in poverty?** (multiple choice). Choices: 1: Under 20% | 2: About 30% | 3: About 40% | 4: Over 50%. Scored correct: About 40% [3].
- **Where was the Democratic Party on the 1–7 scale from lower taxes and services to higher taxes and services?** (placement scale). Choices: 1–7 placement scale; endpoints in question. Scored correct: Positions 5, 6, 7 [5|6|7].
- **Where was the Republican Party on the 1–7 scale from lower taxes and services to higher taxes and services?** (placement scale). Choices: 1–7 placement scale; endpoints in question. Scored correct: Positions 1, 2, 3 [1|2|3].
- **Where was the Democratic Party on the 1–7 scale from government action to reduce unemployment to letting the economy recover?** (placement scale). Choices: 1–7 placement scale; endpoints in question. Scored correct: Positions 1, 2, 3 [1|2|3].
- **Where was the Republican Party on the 1–7 scale from government action to reduce unemployment to letting the economy recover?** (placement scale). Choices: 1–7 placement scale; endpoints in question. Scored correct: Positions 5, 6, 7 [5|6|7].

C.26 California What’s Next (2011)

- **Which party held the majority in the California State Senate?** (multiple choice). Choices: 1: Republican Party | 2: Democratic Party | 98: Don’t Know/ Couldn’t Say. Scored correct: Democratic Party [2].
- **Which party held the majority in the California State Assembly?** (multiple choice). Choices: 1: Republican Party | 2: Democratic Party | 98: Don’t Know/ Couldn’t Say. Scored correct: Democratic Party [2].
- **What legislative majority was needed to approve the proposed state budget?** (multiple choice). Choices: 1: Simple Majority of Both Houses | 2: Simple Majority Assembly & Two-Thirds Senate | 3: Two-Thirds of Both Houses | 4: Three-Fourths of Both Houses | 98: Don’t Know/ Couldn’t Say. Scored correct: Two-Thirds of Both Houses [3].
- **What legislative majority was needed to raise taxes?** (multiple choice). Choices: 1: Simple Majority of Both Houses | 2: Simple Majority Assembly & Two-Thirds Senate | 3: Two-Thirds of Both Houses | 4: Three-Fourths of Both Houses | 98: Don’t Know/ Couldn’t Say. Scored correct: Two-Thirds of Both Houses [3].
- **Who could sign a ballot measure?** (multiple choice). Choices: 1: Anyone 18+ | 2: Any Citizen 18+ | 3: Only Registered Voters | 4: Only Registered Voters Who Voted in Last Election | 98: Don’t Know/ Couldn’t Say. Scored correct: Only Registered Voters [3].

C.27 America in One Room (2019)

- **Which political party holds the majority in the Senate?** (multiple choice). Choices: 1: Democratic | 2: Republican | 3: Independent. Scored correct: Republican [2].
- **Which political party holds the majority in the House?** (multiple choice). Choices: 1: Democratic | 2: Republican | 3: Independent. Scored correct: Democratic [1].
- **About how many undocumented immigrants are in the US?** (multiple choice). Choices: 1: 10 million | 2: 20 million | 3: 30 million | 4: 40 million. Scored correct: 10 million [1].
- **Which of the following countries is NOT part of the Paris Agreement on the environment?** (multiple choice). Choices: 1: Russia | 2: Turkey | 3: Iran | 4: All of the above. Scored correct: All of the above [4].
- **The Affordable Care Act allows which of the following?** (multiple choice). Choices: 1: Stay on your parents' healthcare plan until age 26 | 2: Eliminated annual and lifetime limits on insurance coverage | 3: Cannot charge women higher rates than men | 4: All of the above. Scored correct: All of the above [4].
- **What percentage is the highest tax rate for capital gains taxes?** (multiple choice). Choices: 1: Under 10% | 2: Around 20% | 3: Around 30% | 4: More than 40%. Scored correct: Around 20% [2].
- **Which of the following organizations dealing with trade has the most countries?** (multiple choice). Choices: 1: North American Free Trade Agreement (NAFTA) | 2: World Trade Organization (WTO) | 3: United States-Mexico-Canada Agreement (USMCA) | 4: Trans-Pacific Partnership (TPP). Scored correct: World Trade Organization (WTO) [2].

C.28 America in One Room: Climate (2021)

- **Which political party controls the majority in the Senate?** (multiple choice). Scored correct: Code 1 (answer text not recovered). Offered choice labels were not in the retained report; the keyed code is checked against reported item results.
- **Which political party controls the majority in the House?** (multiple choice). Scored correct: Code 1 (answer text not recovered). Offered choice labels were not in the retained report; the keyed code is checked against reported item results.
- **Which of the following best describes Net Zero?** (multiple choice). Scored correct: Code 3 (answer text not recovered). Offered choice labels were not in the retained report; the keyed code is checked against reported item results.
- **What is the name of a method by which government can encourage emitters to reduce greenhouse gases?** (multiple choice). Scored correct: Code 1 (answer text not recovered). Offered choice labels were not in the retained report; the keyed code is checked against reported item results.
- **Which of the following emit greenhouse gas emissions?** (multiple choice). Scored correct: Code 5 (answer text not recovered). Offered choice labels were not in the retained report; the keyed code is checked against reported item results.

- **Which country emits the most greenhouse gas emissions?** (multiple choice). Scored correct: Code 1 (answer text not recovered). Offered choice labels were not in the retained report; the keyed code is checked against reported item results.
- **The term fossil fuels refers to which of the following energy sources?** (multiple choice). Scored correct: Code 1 (answer text not recovered). Offered choice labels were not in the retained report; the keyed code is checked against reported item results.
- **Which international agreement is about climate change?** (multiple choice). Scored correct: Code 1 (answer text not recovered). Offered choice labels were not in the retained report; the keyed code is checked against reported item results.